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Project Number: #3

Project Name: A Real Application Parallel Challenge

Project Report #3

1. Tell what machine you ran this on

MacBook Air (apple chip M2)

2. Tell what operating system you were using

Mac OS

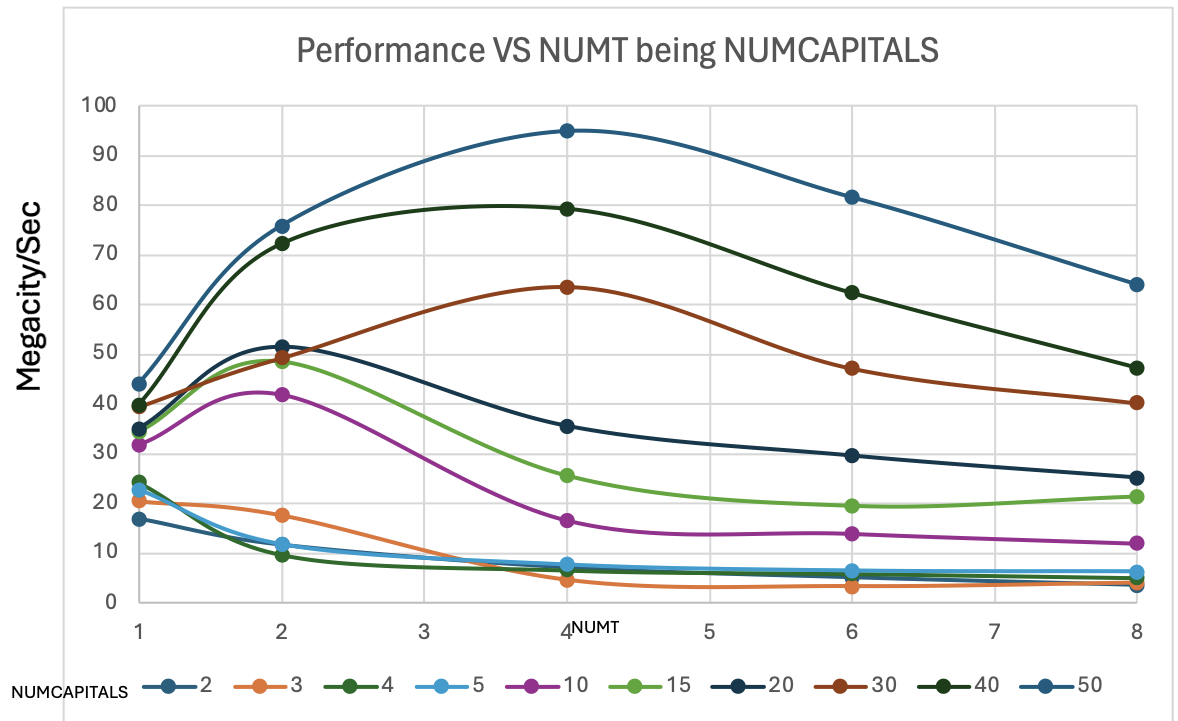
3. Tell what compiler you used

g++

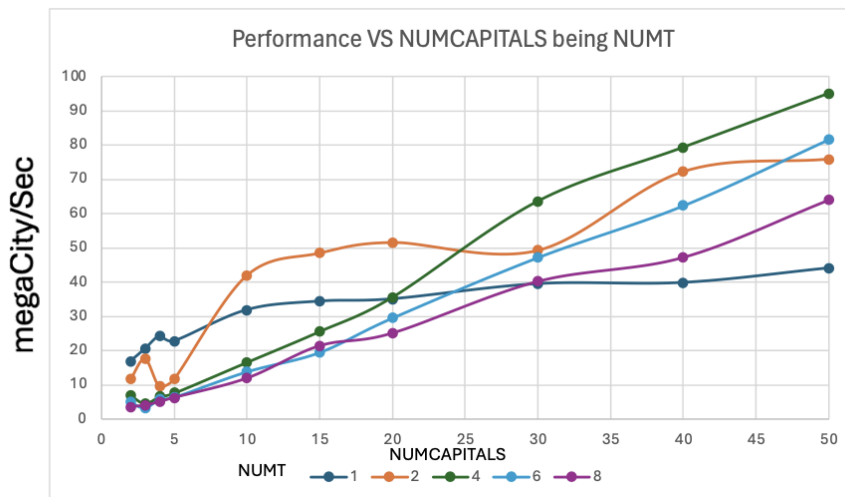
4. Include the table of performance data.

megaCity/Sec	NUMCAPITALS										
NUMT	2	3	4	5	10	15	20	30	40	50	
1	16.877	20.504	24.178	22.81	31.869	34.385	35.056	39.465	39.828	44.047	
2	11.633	17.664	9.681	11.74	41.959	48.493	51.579	49.265	72.223	75.888	
4	7.05	4.618	6.558	7.616	16.586	25.585	35.567	63.578	79.289	94.983	
6	5.052	3.328	5.848	6.4	13.869	19.508	29.532	47.18	62.313	81.6	
8	3.514	4.027	5.027	6.24	11.897	21.369	25.082	40.196	47.232	63.996	

5. Include a graph of performance vs. NUMT with the colored curves being NUMCAPITALS.



6. Include a graph of performance vs. NUMCAPITALS cities with the colored curves being NUMT.



7. Tell us what you discovered by doing this. What patterns are you seeing in the graphs?

Through this experiment, I found that using 4 threads yields the best results(perfoamcne), rather than using just 1 or 8 threads. What I observed is that the performance, which is measured in megacity per second, shows a positive connection with both the number of capitals (NUMCAPITALS) and the number of threads (NUMT). Specifically, regardless of the number of threads used, when we increase the number of capitals, the performance improves. This suggests that having more capitals enhances performance. Similarly, for a fixed number of capitals, performance is better when using a higher number of threads, showing that increasing the number of threads also boosts performance.

8. Extra Credit:

```
● [zouy2@rabbit ~/CS475/project_03$] bash loop.sh
      0:      74.10 ,      40.73 , JerseyCity,NJ
      1:      92.01 ,      41.15 , CedarRapids,IA
      2:     111.39 ,      33.45 , Mesa,AZ
      3:      81.58 ,      28.82 , Orlando,FL
      4:     120.86 ,      45.93 , Gresham,OR
      5:     117.77 ,      34.11 , Pomona,CA
      6:      96.70 ,      31.77 , Waco,TX
      7:     106.77 ,      39.74 , Boulder,CO
      8:      83.57 ,      37.95 , Lexington,KY
      9:     121.65 ,      37.98 , Antioch,CA
```